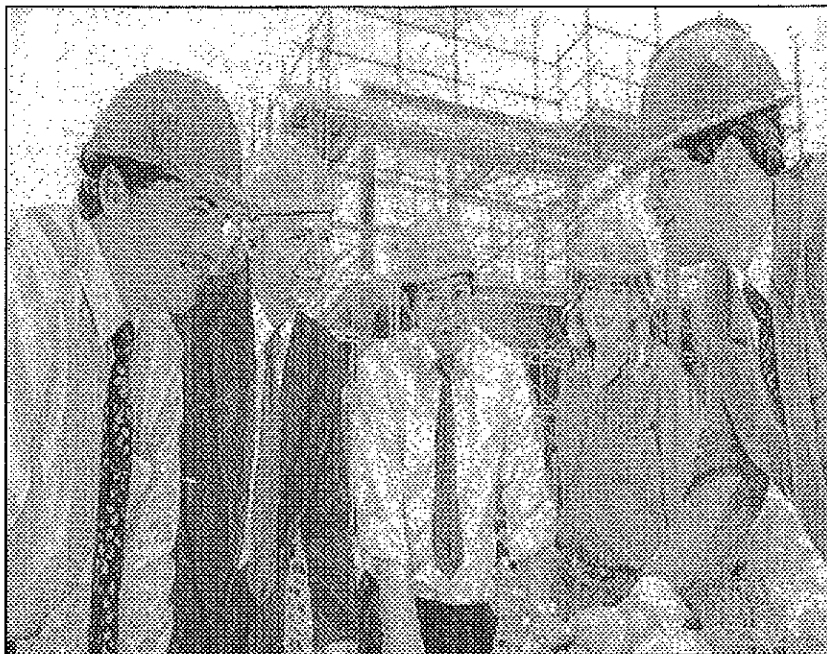


SPECIFICATIONS FOR YORKSHIRE ELECTRICITY DESIGN LINES TO
CE/C/37(M1)

MAJOR PROJECTS

Distribution



JOHN COOK

DESIGN ENGINEER



3. SPECIFICATIONS FOR YORKSHIRE ELECTRICITY DESIGN LINES TO CE/C/37(M1) & CE/OM/DC 1(M)

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	<u>Drawings</u>	<u>Drawing No</u>
	CE/C/37(M1) Intermediate	Y104 L 0112
	CE/C/37(M1) Section	Y104 L 0113
	CE/C/37(M1) Angle up to 10°	Y104 L 0114
	CE/C/37(M1) Angle 10° to 60°	Y104 L 0115
	CE/C/37(M1) Slack Span Terminal	Y104 L 0116
	CE/C/37(M1) or CE/OM/DC 1(M) 33kV Cable Termination(Sealing End) with S.Ds.	Y104 L 0117
	CE/C/37(M1) or CE/OM/DC 1(M) 66kV Cable Termination (Sealing End) with S.Ds. (short poles)	Y104 L 0118
	CE/C/37(M1) or CE/OM/DC 1(M) 66kV Cable Termination (Sealing End) with S.Ds. (tall poles)	Y104 L 0108
	CE/C/37(M1) or CE/OM/DC 1(M) 33kV Polymeric Cable Termination with S.Ds.	Y104 L 0109
	CE/C/37(M1) or CE/OM/DC 1(M) 66kV Polymeric Cable Termination with S.Ds.	Y104 L 0111
	CE/OM/DC 1(M) Intermediate	Y105 L 0101
	CE/OM/DC 1(M) Section	Y105 L 0102
	CE/OM/DC 1(M) Angle	Y105 L 0103

Revision 7 (May 1998)

3 SPECIFICATIONS FOR YORKSHIRE ELECTRICITY DESIGN LINES TO CE/C/37(M1) AND CE/OM/DC 1(M).

3.1.WEATHER LOCATIONS.

Lines constructed to these specifications are suitable for use in all locations within Yorkshire Electricity's area.

3.2.CONDUCTOR.

Lines shall be constructed using Aluminium Conductor Steel Reinforced, 175sq.mm.(37/2.79mm.) code name LYNX or 200sq mm (37/2.87mm High Conductive) All Aluminum Alloy Conductor code name POPLAR.

3.3.SPAN LENGTHS.

Circuits shall be planned such that the average span length of each self supporting section does not exceed the Recommended Span.

For CE/C/37(M1) lines longer spans may be obtained by taking advantage of the increased span lengths detailed in the Wind Loading Span table.

3.4.CONDUCTOR ERECTION.

Final conductor sags and tensions shall be determined from the Design Sag and Tension Table for each specification.

The erection tension for new conductors shall be determined from the Erection Tension Table for each specification.

Where existing conductors are to be re-used the erection tension shall be determined from the Design Sag and Tension Table for each specification.¹

3.5.SUPPORTS.

Supports shall be assembled to the following drawings for CE/C/37(M1):-

DRAWING NUMBER	SUPPORT TYPE
Y104 L 0112	Intermediate
Y104 L 0113	Section
Y104 L 0114	Angle < 10°
Y104 L 0115	Angle 10° - 60°
Y104 L 0116	Slack Span Terminal
Y104 L 0117	33kV Cable Terminal (SE) with S.D.
Y104 L 0118	66kV Cable Terminal (SE) with S.D.
Y104 L 0108	66kV Cable Term (SE) with S.D.(tall poles)
Y104 L 0109	33kV Polymeric Cable Term with S.D.
Y104 L 0111	66kV Polymeric Cable Term with S.D.

and to the following drawings for CE/OM/DC 1(M):-

DRAWING NUMBER	SUPPORT TYPE
Y105 L 0101	Intermediate
Y105 L 0102	Section
Y105 L 0103	Angle

Termination of CE/OM/DC 1(M) shall be on CE/C/37(M1) structures.

¹ Revision 7 (May 1998)

3.6.STAYS.

All staywires shall be set at 45° to the pole as determined by the specifications.

Stay insulators shall be installed in accordance with the following table :-

VOLTAGE	EARTHED	UNEARTHED
	SUPPORT	SUPPORT
33kV	1 Type 1 P	2 Type 2 P
66kV	1 Type 1 P	1 Beam Type

NOTE! P = Porcelain.

3.7.

CE/C/37(M1)

175sq.mm ACSR and 200sq.mm AAAC ALL LEVELS.

Recommended Span. 150m.

Wind Loading Span (m)	167	163	159	161	171	187	180	197	205
Support Length (m)	10.5	11.0	11.5	12.0	13.0	14.0	15.0	17.0	18.0

IN LINE STRUCTURES.

DRAWING NUMBER	SUPPORT TYPE	SUPPORT CLASS
Y104 L 0112	H Inter	Stout
Y104 L 0113	H Section	Stout

ANGLE STRUCTURES.

DRAWING NUMBER	SUPPORT TYPE	SUPPORT CLASS	LINE DEVIATION:	STAYS Nº
Y104 L 0114	H Sect Angle	Stout	10°	1 at 45°
Y104 L 0114	H Sect Angle	Stout	15°	1 at 45°
Y104 L 0115	H Sect Angle	Stout	46°	3 at 45°
Y104 L 0115	H Sect Angle	Stout	60°	4 at 45°

Stay Positions.

On 1 stay installations the stay is fixed to the outer limb of the structure.

On 3 stay installations 1 stay is fixed to the outer limb and 2 stays to the inner limb

On 4 stay installations 2 stays are fixed to the inner limb and 2 stays are fixed to the outer limb.

Revision 7 (May 1998)

TERMINAL STRUCTURES.

DRAWING NUMBER	SUPPORT TYPE	SUPPORT CLASS	STAYS Nº
Y104 L 0116	Slack Span Term (33/66kV)	Stout	2/leg
Y104 L 0109	Cable Term (33kV)	Stout	2/leg
Y104 L 0111	Cable Term (66kV)	Stout	2/leg

SAG TABLES

Conductor design sags and tensions are given in tables 3.7A & 3.7A1

Conductor erection sags and tensions are given in tables 3.7B
in The Overhead Line Manual Part 2 for 175 ACSR.

For 200sq.mm AAAC Erection sags and tensions are the same as Design sags
and tensions.

Revision 7 (May 1998)

175sq.mm.ACSR
CE/C/37(M1)

ALL LEVELS.
150m. BASIC SPAN

Conductor Breaking Load. 79800 Newtons.
Conductor Diameter. 19.53mm.
Conductor Weight. 0.8628 kg/m.
Conductor c.s.a. 226.2 sq.mm.
Maximum Working Tension. 25740 Newtons.

Span Length (m)	Temperature (deg C)															
	0	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75
60	0.23	0.25	0.27	0.28	0.30	0.32	0.33	0.35	0.37	0.39	0.41	0.42	0.44	0.46	0.47	0.49
65	0.27	0.29	0.31	0.33	0.35	0.37	0.39	0.41	0.43	0.46	0.48	0.50	0.52	0.54	0.56	0.58
70	0.32	0.34	0.36	0.38	0.41	0.43	0.46	0.48	0.50	0.53	0.55	0.58	0.60	0.62	0.65	0.67
75	0.37	0.39	0.41	0.44	0.47	0.49	0.52	0.55	0.58	0.61	0.63	0.66	0.69	0.72	0.74	0.77
80	0.42	0.44	0.47	0.50	0.53	0.56	0.59	0.63	0.66	0.69	0.72	0.75	0.78	0.81	0.84	0.87
85	0.47	0.50	0.53	0.57	0.60	0.64	0.67	0.71	0.74	0.78	0.81	0.85	0.88	0.92	0.95	0.99
90	0.53	0.56	0.60	0.63	0.67	0.71	0.75	0.79	0.83	0.87	0.91	0.95	0.99	1.03	1.07	1.10
95	0.59	0.63	0.67	0.71	0.75	0.79	0.84	0.88	0.93	0.97	1.02	1.06	1.10	1.15	1.19	1.23
100	0.65	0.69	0.74	0.78	0.83	0.88	0.93	0.98	1.03	1.08	1.13	1.18	1.22	1.27	1.32	1.36
105	0.72	0.76	0.81	0.86	0.92	0.97	1.02	1.08	1.13	1.19	1.24	1.30	1.35	1.40	1.45	1.50
110	0.79	0.84	0.89	0.95	1.01	1.06	1.12	1.18	1.25	1.30	1.36	1.42	1.48	1.54	1.60	1.65
115	0.86	0.92	0.97	1.04	1.10	1.16	1.23	1.29	1.36	1.43	1.49	1.56	1.62	1.68	1.74	1.80
120	0.94	1.00	1.06	1.13	1.20	1.27	1.34	1.41	1.48	1.55	1.62	1.69	1.76	1.83	1.90	1.96
125	1.02	1.08	1.15	1.22	1.30	1.37	1.45	1.53	1.61	1.69	1.76	1.84	1.91	1.99	2.06	2.13
130	1.10	1.17	1.25	1.32	1.40	1.49	1.57	1.65	1.74	1.82	1.90	1.99	2.07	2.15	2.23	2.30
135	1.19	1.26	1.34	1.43	1.51	1.60	1.69	1.78	1.88	1.97	2.05	2.14	2.23	2.32	2.40	2.48
140	1.27	1.36	1.44	1.54	1.63	1.72	1.82	1.92	2.02	2.11	2.21	2.31	2.40	2.49	2.58	2.67
145	1.37	1.46	1.55	1.65	1.75	1.85	1.95	2.06	2.16	2.27	2.37	2.47	2.57	2.67	2.77	2.87
150	1.46	1.56	1.66	1.76	1.87	1.98	2.09	2.20	2.32	2.43	2.54	2.65	2.75	2.86	2.97	3.07
155	1.56	1.66	1.77	1.88	2.00	2.11	2.23	2.35	2.47	2.59	2.71	2.83	2.94	3.06	3.17	3.28
160	1.67	1.77	1.89	2.01	2.13	2.25	2.38	2.51	2.63	2.76	2.89	3.01	3.13	3.26	3.38	3.49
165	1.77	1.89	2.01	2.13	2.26	2.39	2.53	2.66	2.80	2.94	3.07	3.20	3.33	3.46	3.59	3.71
170	1.88	2.00	2.13	2.26	2.40	2.54	2.68	2.83	2.97	3.12	3.26	3.40	3.54	3.68	3.81	3.94
175	1.99	2.12	2.26	2.40	2.54	2.69	2.84	3.00	3.15	3.30	3.45	3.60	3.75	3.89	4.04	4.18
180	2.11	2.24	2.39	2.54	2.69	2.85	3.01	3.17	3.33	3.49	3.65	3.81	3.97	4.12	4.27	4.42

Tension (kgf) 1658 1557 1463 1377 1298 1226 1161 1102 1048 1000 957 917 881 848 818 791

TABLE 3.7A DESIGN SAGS AND TENSIONS

200sq.mm.AAAC (Poplar)
CE/C/37(M1)

ALL LEVELS.
150m.BASIC SPAN

Conductor Breaking Load. 67090 Newtons.
 Conductor Diameter. 20.09 mm.
 Conductor Weight. 0.659 kg/m.
 Conductor c.s.a. 200 sq.mm.
 Maximum Working Tension. 23514 Newtons.

Span Length (m)	Temperature (deg C)										
	0	5	10	15	20	25	30	35	50	60	65
60	0.21	0.22	0.24	0.26	0.28	0.30	0.33	0.35	0.42	0.46	0.48
65	0.24	0.26	0.28	0.31	0.33	0.36	0.38	0.41	0.49	0.54	0.57
70	0.28	0.31	0.33	0.36	0.38	0.41	0.44	0.48	0.57	0.63	0.66
75	0.32	0.35	0.38	0.41	0.44	0.48	0.51	0.55	0.65	0.72	0.75
80	0.37	0.40	0.43	0.47	0.50	0.54	0.58	0.62	0.74	0.82	0.86
85	0.42	0.45	0.49	0.53	0.57	0.61	0.66	0.70	0.84	0.92	0.97
90	0.47	0.50	0.55	0.59	0.64	0.68	0.74	0.79	0.94	1.04	1.09
95	0.52	0.56	0.61	0.66	0.71	0.76	0.82	0.88	1.05	1.16	1.21
100	0.58	0.62	0.67	0.73	0.79	0.85	0.91	0.97	1.16	1.28	1.34
105	0.63	0.69	0.74	0.80	0.87	0.93	1.00	1.07	1.28	1.41	1.48
110	0.70	0.75	0.81	0.88	0.95	1.02	1.10	1.17	1.40	1.55	1.62
115	0.76	0.82	0.89	0.96	1.04	1.12	1.20	1.28	1.53	1.69	1.77
120	0.83	0.90	0.97	1.05	1.13	1.22	1.31	1.40	1.67	1.84	1.93
125	0.90	0.97	1.05	1.14	1.23	1.32	1.42	1.51	1.81	2.00	2.09
130	0.97	1.05	1.14	1.23	1.33	1.43	1.53	1.64	1.96	2.16	2.27
135	1.05	1.13	1.23	1.33	1.43	1.54	1.65	1.77	2.11	2.33	2.44
140	1.13	1.22	1.32	1.43	1.54	1.66	1.78	1.90	2.27	2.51	2.63
145	1.21	1.31	1.42	1.53	1.65	1.78	1.91	2.04	2.44	2.69	2.82
150	1.30	1.40	1.52	1.64	1.77	1.90	2.04	2.18	2.61	2.88	3.02
155	1.38	1.50	1.62	1.75	1.89	2.03	2.18	2.33	2.78	3.08	3.22
160	1.47	1.59	1.72	1.86	2.01	2.16	2.32	2.48	2.96	3.28	3.43
165	1.57	1.70	1.83	1.98	2.14	2.30	2.47	2.64	3.15	3.49	3.65
170	1.66	1.80	1.95	2.10	2.27	2.44	2.62	2.80	3.35	3.70	3.87
175	1.76	1.91	2.06	2.23	2.40	2.59	2.78	2.97	3.55	3.92	4.11
180	1.87	2.02	2.18	2.36	2.54	2.74	2.94	3.14	3.75	4.15	4.34
Tension (kgf)	1478	1367	1264	1169	1084	1007	938	878	735	665	635

TABLE 3.7A1 DESIGN SAGS AND TENSIONS

MATERIALS LIST.

Item	Nº	Commodity Code	Description	Drawing or Ref.Number	M 20 Bolts	Washers		
						sc	sf	r
3	2	BPC 024	Wood Pole (type 1A)	Y204 L 0601				
29	1	222844	Crossarm Member	Y204 L 0901	2(Pole)	2		
58	3		33kV Suspension Set	Y204 L 0703				
			66kV Suspension Set					
70	2	294181	Safety Sign	Y201 L 1002				
71			Operational Plates	Y201 L 1001/7				
45	1	219519	Pole Bracing (Single)	Y204 L 0905				
		219501	Pole Bracing (Double)	0301				
19	2	217497	Brace Block	Y204 L 1001/7	4(Pole)	4	4	

Notes to drawing N° Y104 L 0112.

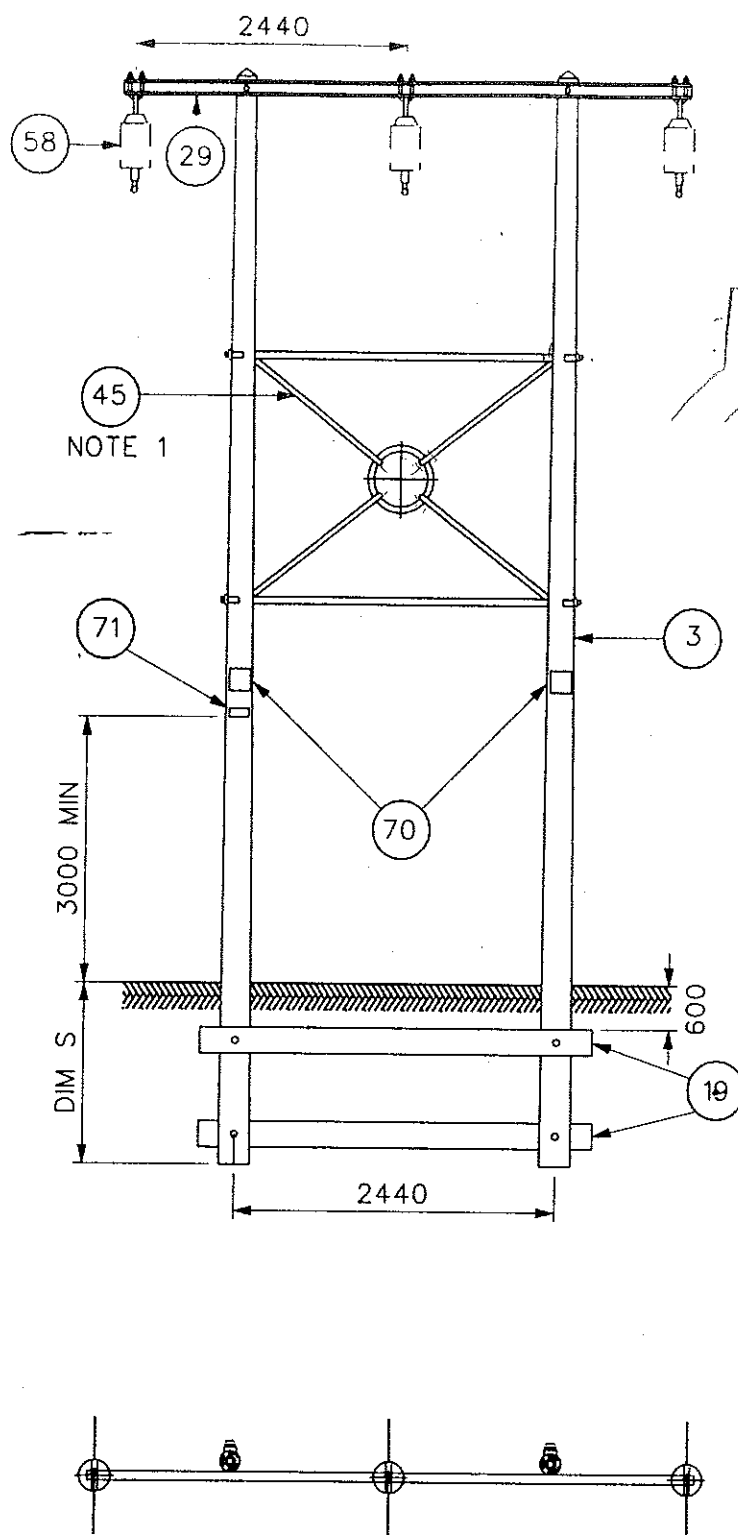
1. Single pole bracing required for poles up to 14.0m.

Double pole bracing required for poles 15.0m and above.

Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=11.5	1.80
12.0 - 13.0	2.00
14.0 - 15.0	2.10
16.0 - 17.0	2.25
18.0 - 20.0	2.40

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DIMENSIONS IN MILLIMETRES

drawn	GR	CE/C/37 (M1) INTERMEDIATE			Y104L0112	CAD	 Yorkshire Electricity
traced	RLH						
checked							
approved							
revision	A	date	18.5.98	original scale	N.T.S		

MATERIALS LIST.

Item	Nº	Commodity Code	Description	Drawing or Ref. Number	M 20 Bolts	Washers		
						sc	sf	r
3	2	BPC 024	Wood Pole (type 1A)	Y204 L 0601				
30	1	222455	Crossarm Member	Y204 L 0902	2 x 300			
60	6		33kV Tension Set	Y204 L 0704				
			66kV Tension Set					
70	2	294181	Safety Sign	Y201 L 1002				
71			Operational Plates	Y201 L 1001/7				
45	1	219519	Pole Bracing (Single)	Y 204 L 0905				
		219501	Pole Bracing (Double)					
19	2	217497	Brace Block	Y204 L 0301	4(Pole)	4	4	

Notes to drawing Nº Y104 L 0113.

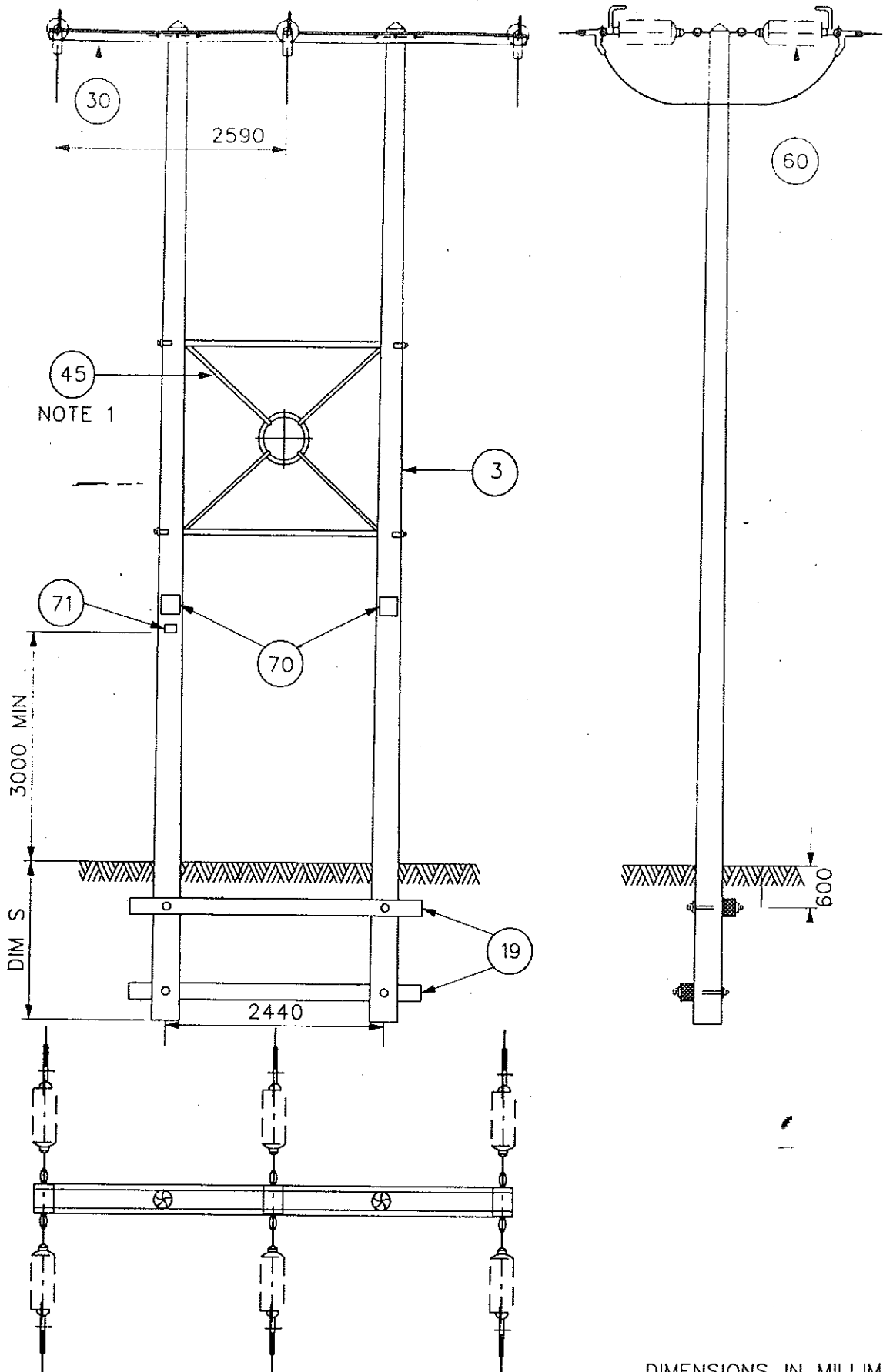
1. Single pole bracing required for poles up to 14.0m.

Double pole bracing required for poles 15.0m and above.

Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=11.5	1.80
12.0 - 13.0	2.00
14.0 - 15.0	2.10
16.0 - 17.0	2.25
18.0 - 20.0	2.40

Area:
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DIMENSIONS IN MILLIMETRES

drawn	GR	CE/C/37 (M1) SECTION			Y104L0113	CAD	 Yorkshire Electricity
traced	RLH						
checked							
approved							
revision	B	date	18.5.98	original scale	NTS		

MATERIALS LIST.

Item	Nº	Commodity Code	Description	Drawing or Ref. Number	M 20 Bolts	Washers		
						sc	sf	r
4	2	BPC 024	Wood Pole (type 2)	Y204 L 0601				
19	2	217497	Brace Block	Y204 L 0301	4(Pole)	4	4	
30	1	222455	Crossarm Member	Y204 L 0902	4 x 300*			
45	1	219519	Pole Bracing(single)	Y204 L 0905				
		219501	Pole Bracing(double)					
60	6		33kV Tension Set	Y204 L 0704				
			66kV Tension Set					
70	2	294181	Safety Sign	Y201 L 1002				
71			Operational Plates	Y201 L 1001/7				
73			Stay Arrangement	Y204 L 0501				

* 8 off M20 x 45 bolts also required for gusset plates.

Notes to drawing Nº Y104 L 0114.

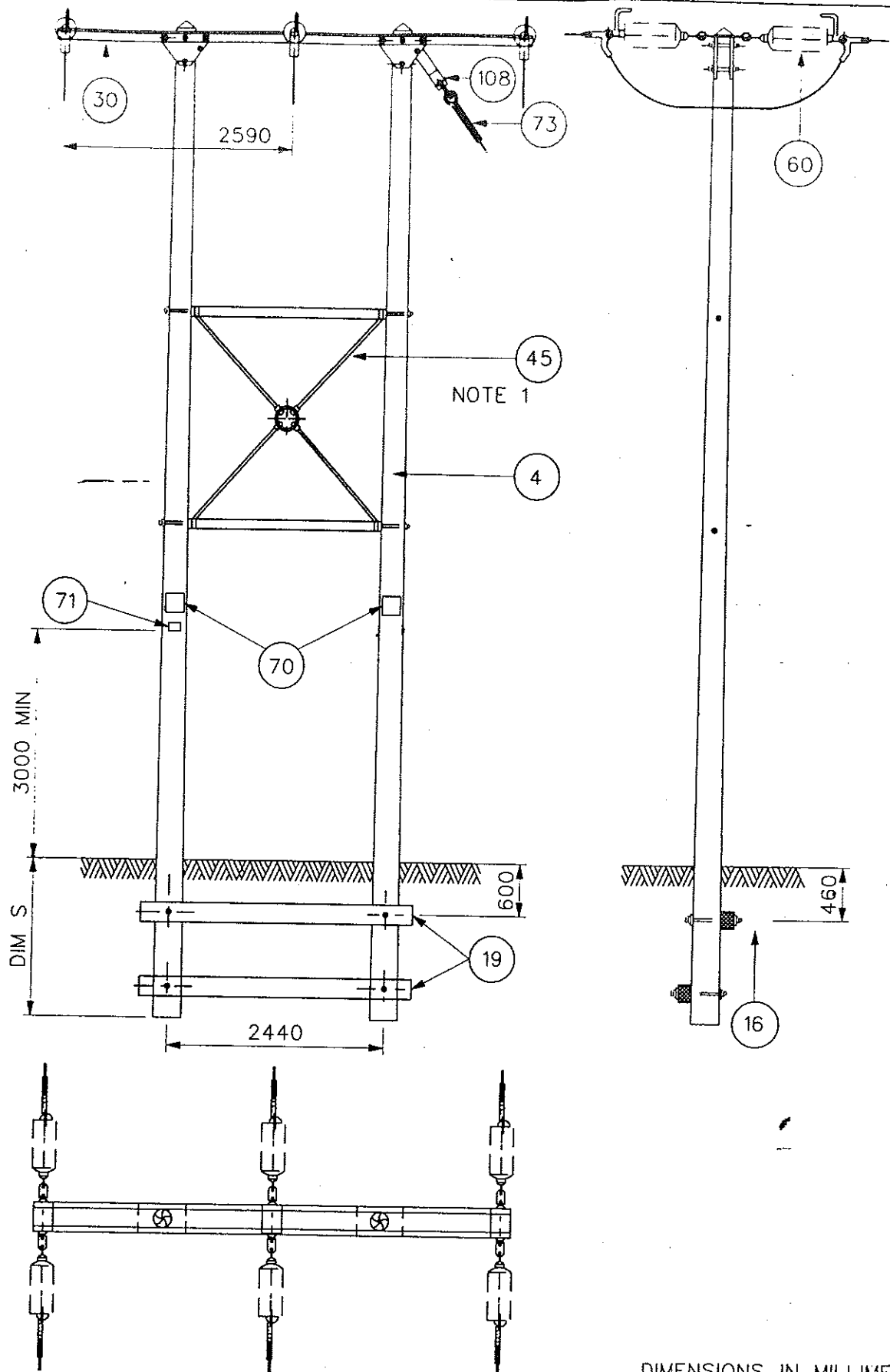
1. Single pole bracing required for poles up to 14.0m.

Double pole bracing required for poles 15.0m and above.

Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=11.5	1.80
12.0 - 13.0	2.00
14.0 - 15.0	2.10
16.0 - 17.0	2.25
18.0 - 20.0	2.40

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DIMENSIONS IN MILLIMETRES

drawn	GR	CE/C/37 (M1) ANGLE UP TO 10°		
traced	RLH			
checked				
approved				
revision	A	date	18.5.98	original scale
			NTS	Y104L0114

CAD



MATERIALS LIST.

Item	Nº	Commodity Code	Description	Drawing or Ref.Number	M 20 Bolts	Washers		
						sc	sf	r
4	2	BPC 024	Wood Pole (type 2)	Y204 L 0601				
19	2	217497	Brace Block	Y204 L 0301	4(Pole)	4	4	
31	1	222463	Crossarm Member	Y204 L 0903	4 x 300			
45	1	219519	Pole Bracing(single)	Y204 L 0905				
		219501	Pole Bracing(double)					
51	1							
		225763	Pilot Pin	BS 3288 ref 50				
55	1	216069	33kV Pin Insulator	439302				
56	1	215830	33kV Line Post Insulator	Y204 L 0701				
		215863	66kV Line Post Insulator					
60	6		33kV Tension Set	Y204 L 0704				
			66kV Tension Set					
70	2	294181	Safety Sign	Y201 L 1002				
71			Operational Plates	Y201 L 1001/7				
73			Stay Arrangement	Y204 L 0501				

* 8 off M20 x 45 bolts also required for gusset plates.

Notes to drawing N° Y104 L 0115.

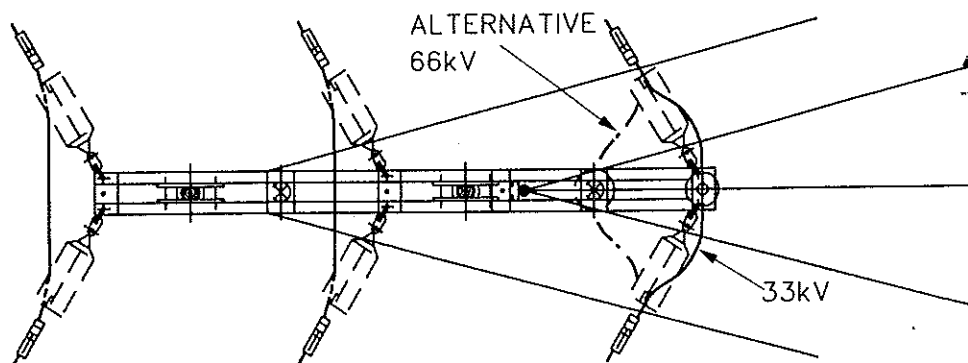
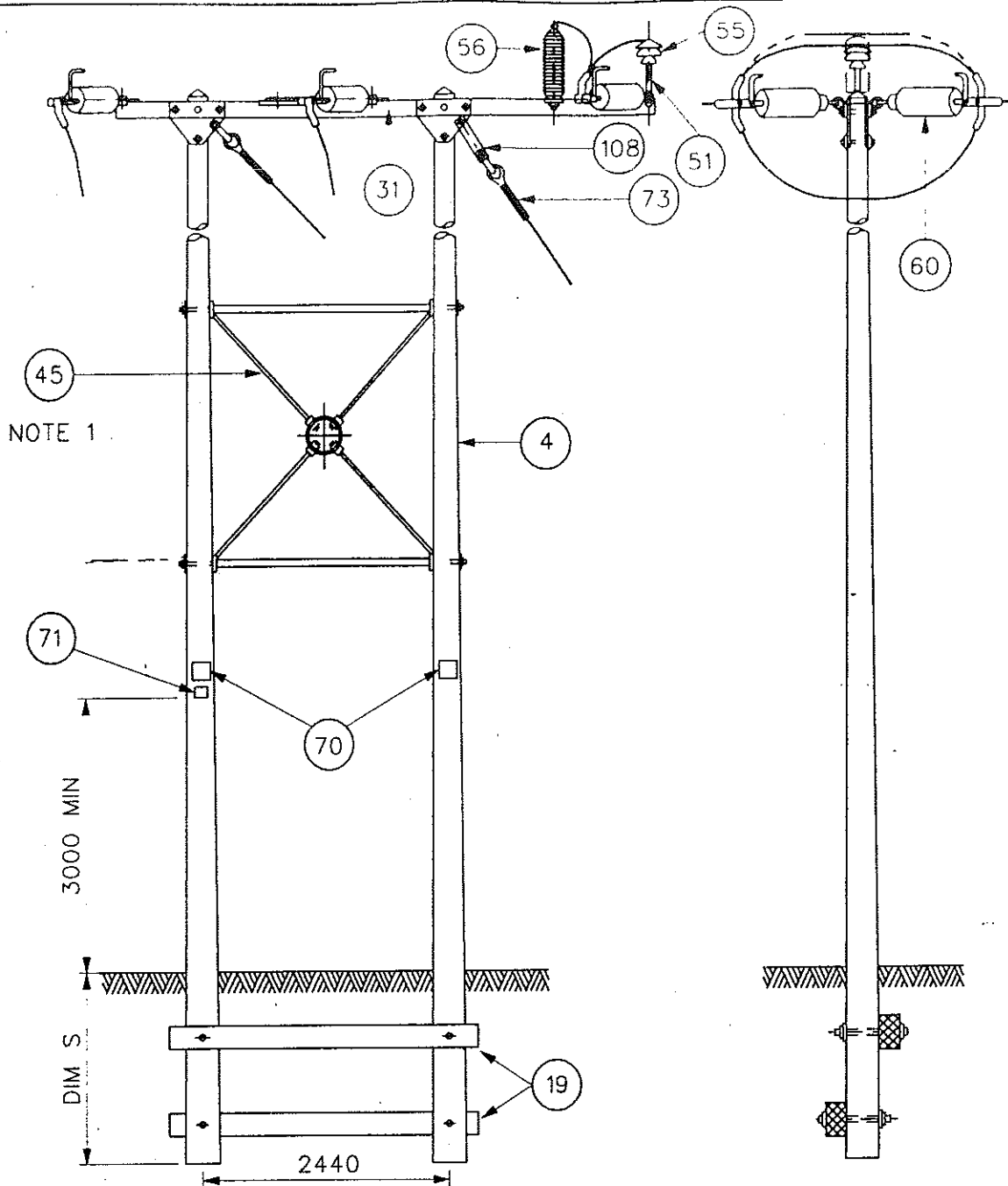
1. Single pole bracing required for poles up to 14.0m.

Double pole bracing required for poles 15.0m and above.

Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=11.5	1.80
12.0 - 13.0	2.00
14.0 - 15.0	2.10
16.0 - 17.0	2.25
18.0 - 20.0	2.40

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DIMENSIONS IN MILLIMETRES

drawn	BT	CE/C/37 (M1) ANGLE 10° TO 60°			
traced	SW				
checked	JMP				
approved					
revision	B	date	18.5.98	original scale	NTS
				Y104L0115	
				CAD	



Yorkshire
Electricity

MATERIALS LIST.

Item	Nº	Commodity Code	Description	Drawing or Ref.Number	M 20 Bolts	Washers		
						sc	sf	r
5	2	BPC 024	Wood Pole (type 3)	Y204 L 0601				
19	2	217497	Brace Block	Y204 L 0301	4(Pole)	4	4	
32	1	223669	Crossarm Member	Y204 L 0904	4 x 300			
45	1	219519	Pole Bracing(single)	Y204 L 0905				
		219501	Pole Bracing(double)					
60	9		33kV Tension Set	Y204 L 0704				
			66kV Tension Set					
70	2	294181	Safety Sign	Y201 L 1002				
71			Operational Plates	Y201 L 1001/7				
73	4		Stay Arrangement	Y204 L 0501				

* 8 off M20 x 45 bolts also required for gusset plates.

Notes to drawing Nº Y104 L 0116.

1. Single pole bracing required for poles up to 14.0m.

Double pole bracing required for poles 15.0m and above.

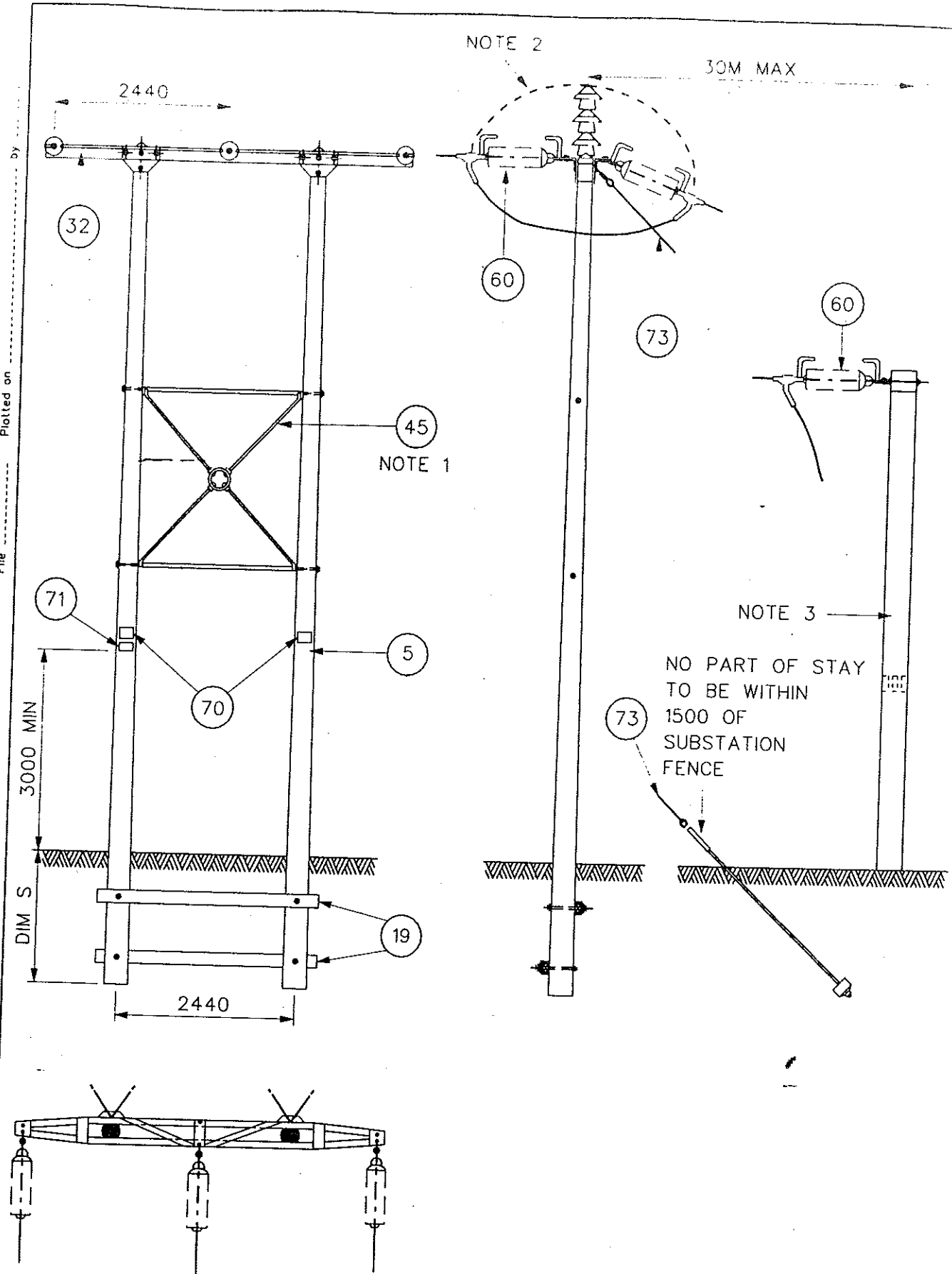
2. Alternative jumper position using 66kV post insulator, or 33kV pin or line post insulator, to be used where adequate clearances cannot be obtained with underslung jumper position.

3. Arrangement of concrete slack span support to drawing number Y003 S 3307.

Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=11.5	1.80
12.0 - 13.0	2.00
14.0 - 15.0	2.10
16.0 - 17.0	2.25
18.0 - 20.0	2.40

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Area: File: Plotted on: by:



DIMENSIONS IN MILLIMETRES

drawn	BT	CE/C/37 (M1) SLACK SPAN TERMINAL		
traced	AJR			
checked	JMP			
approved				
revision	A	date 18-5-98	original scale NTS	Y104L0116



Yorkshire Electricity



MATERIALS LIST.

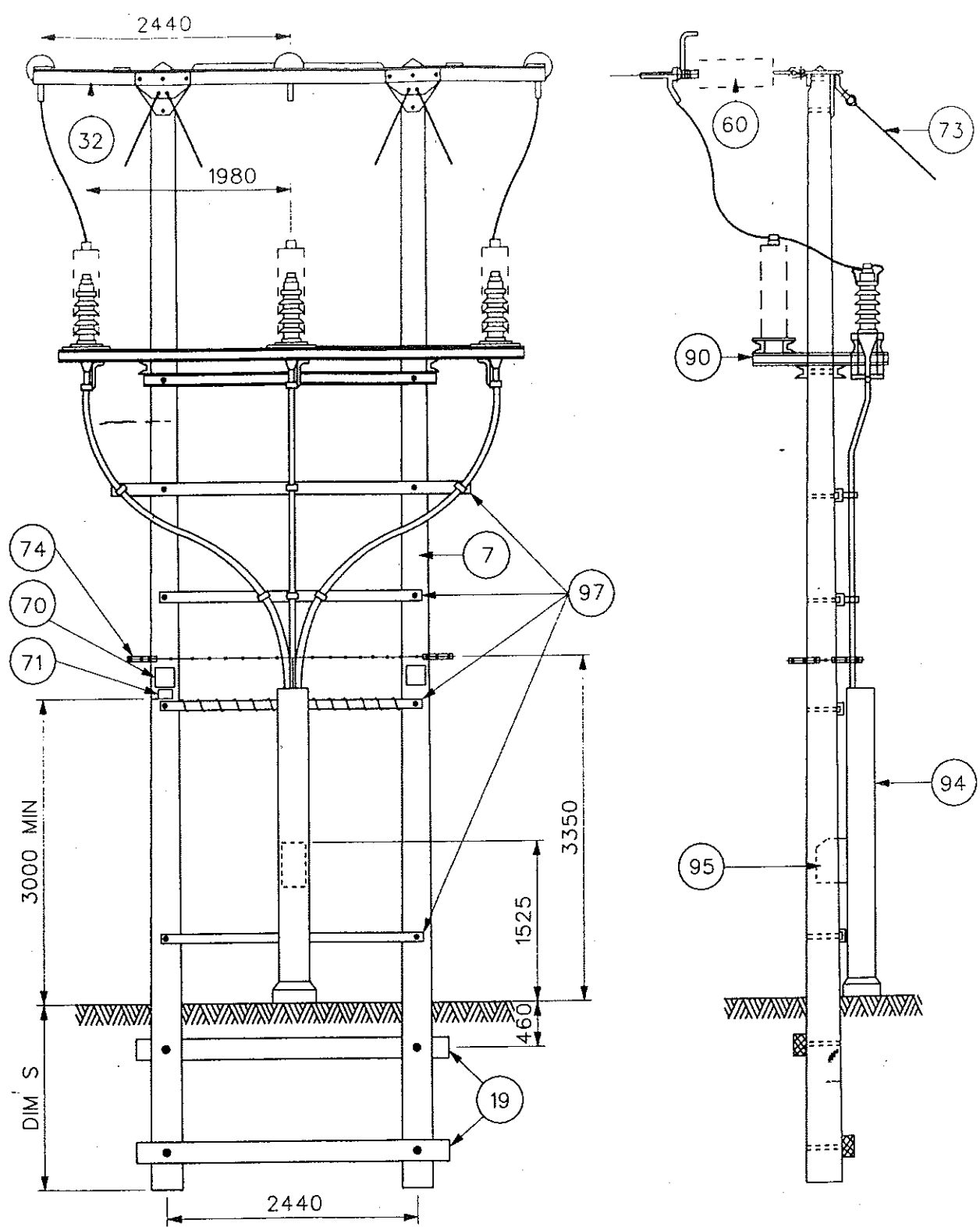
Item	Nº	Commodity Code	Description	Drawing or Ref.Number	M 20 Bolts	Washers		
						sc	sf	r
7	2	BPC 024	Wood Pole	Y204 L 0602				
19	2	217497	Brace Block	Y204 L 0301	4(Pole)	4	4	
32	1	223669	Crossarm Member	Y204 L 0904	4 x 300'			
60	3		33kV Tension Set	Y204 L 0704				
			66kV Tension Set					
70	2	294181	Safety Sign	Y201 L 1002				
71			Operational Plates	Y201 L 1001/7				
73	4		Stay Arrangement	Y204 L 0501				
74	1	224519	Anti- Climbing Guard	Y201 L 1101/2				
90	1	227173	33kV Sealing End P-form	Y204 L 0907	10(Pole)	10		
94	1	224535	Cable Guard	Y204 L 0911	4 x 50			
95	1		Link Box	Eng Rec C55				
97	1		33kV Cable Supports	Y204 L 0915	8(Pole)			

* 4 off M20 x 60 bolts also required for gusset plates.

Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=11.5	1.80
12.0 - 13.0	2.00
14.0 - 15.0	2.10
16.0 - 17.0	2.25
18.0 - 20.0	2.40

Area:
 File: Plotted on: by:

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DIMENSIONS IN MILLIMETRES

drawn	BT	GENERAL ARRANGEMENT		
traced	IPM	CE/C/37 (M1) CE/OM/DC1 33kV CABLE TERMINATION WITH SURGE DIVERTERS		
checked	JMP			
approved				
revision	B	date	18.5.98	original scale NTS
		Y104L0117		CAD



Yorkshire
Electricity

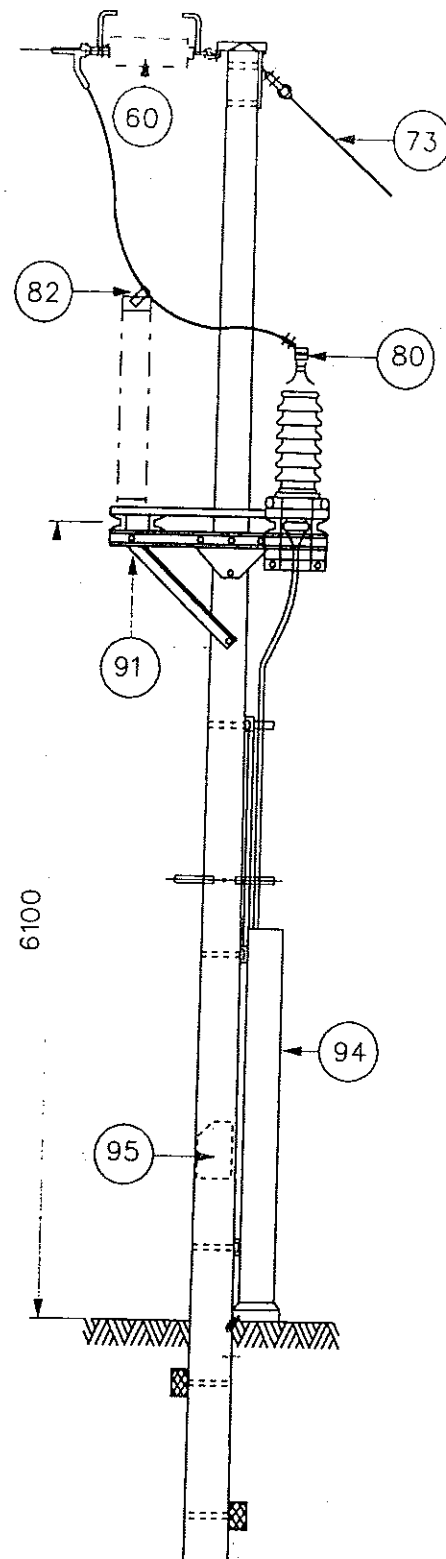
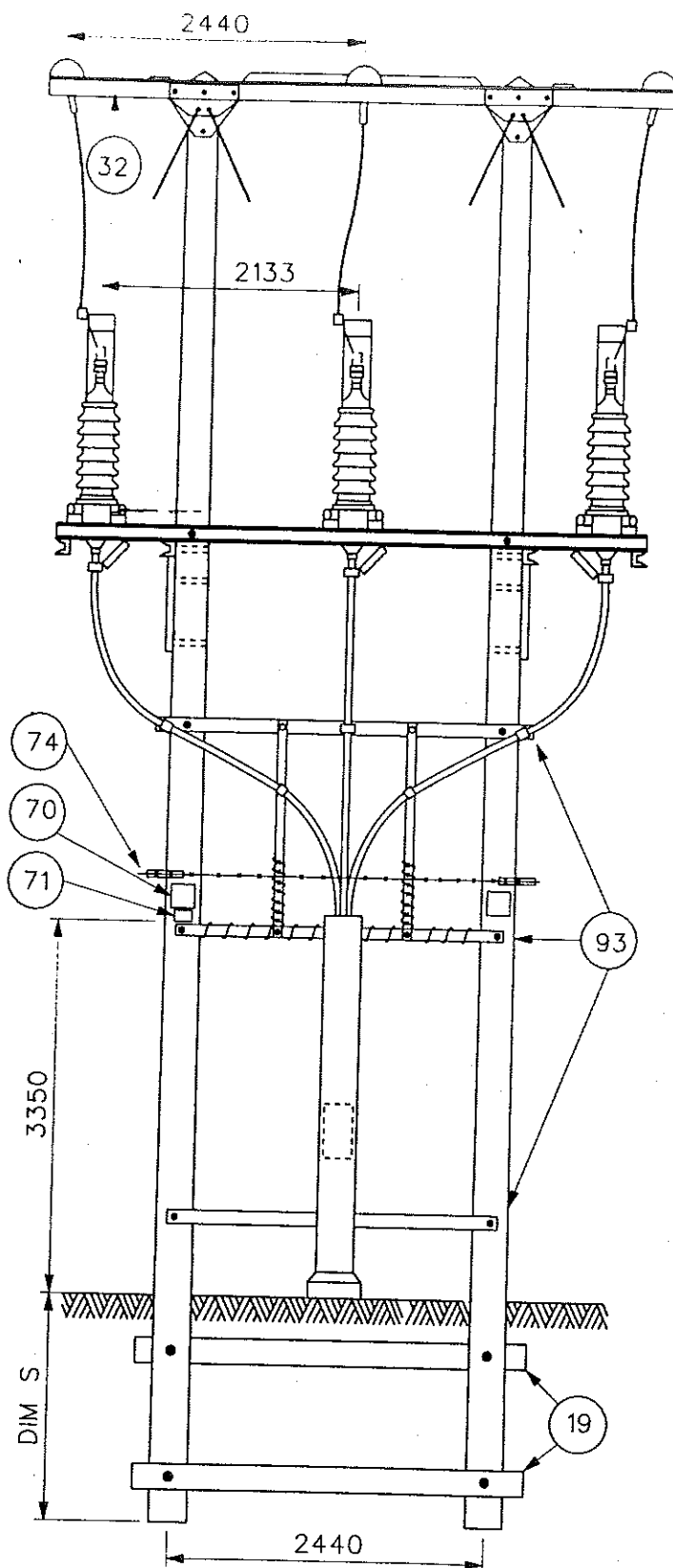
MATERIALS LIST.

Item	Nº	Commodity Code	Description	Drawing or Ref. Number	M 20 Bolts	Washers		
8	2	BPC 024	Wood Pole	Y204 L 0603		sc	sf	r
19	2	217497	Brace Block	Y204 L 0301	4(Pole)	4	4	
32	1	223669	Crossarm Member	Y204 L 0904	4 x 300			
60	3		33kV Tension Set	Y204 L 0704				
			66kV Tension Set					
70	2	294181	Safety Sign	Y201 L 1002				
71			Operational Plates	Y201 L 1001/7				
73	4		Stay Arrangement	Y204 L 0501				
74	1	224519	Anti- Climbing Guard	Y201 L 1101/2				
80	3		66kV Sealing End Connector	Y204 L 0801				
82	3	221119	Surge Diverter Connector	Y204 L 0803				
91	1	227116	66kV Sealing End P-form	Y204 L 0908	6(Pole)			
93	1		66kv Cable Supports	Y204 L 0910	6(Pole)			
94	1	224535	Cable Guard	Y204 L 0911	4 x 50			
95	1		Link Box	Eng Rec C55				

* 4 off M20 x 60 bolts also required for gusset plates.

Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=11.5	1.80
12.0 - 13.0	2.00
14.0 - 15.0	2.10
16.0 - 17.0	2.25
18.0 - 20.0	2.40

Area: _____
 File: _____
 Plotted on: _____ by: _____



DIMENSIONS IN MILLIMETRES

drawn	BT
traced	JEC
checked	JMG
approved	
revision	A

CE/C/37 (M1) CE/OM/DC1 66kV CABLE TERMINATION WITH SURGE DIVERTERS 11.5 TO 14 OM POLES		
date	18.5.98	original scale
	NTS	Y104L0118



Yorkshire
Electricity



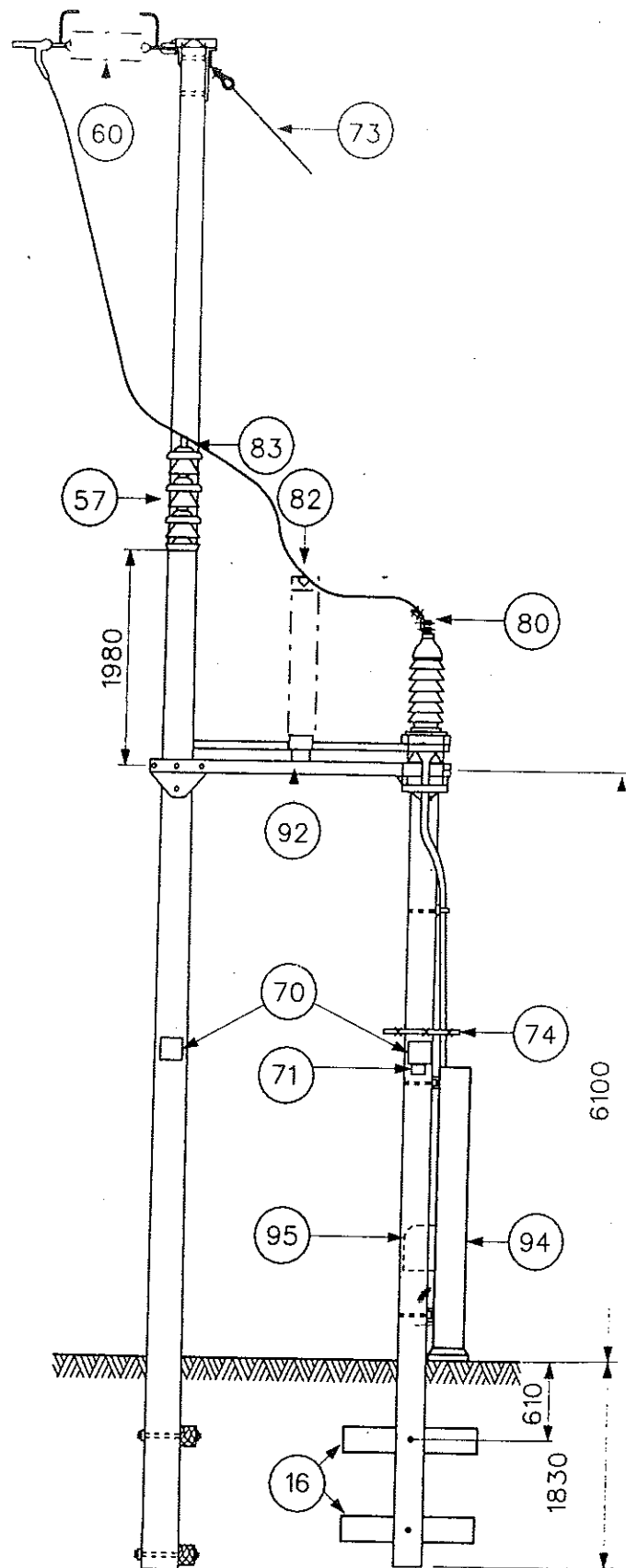
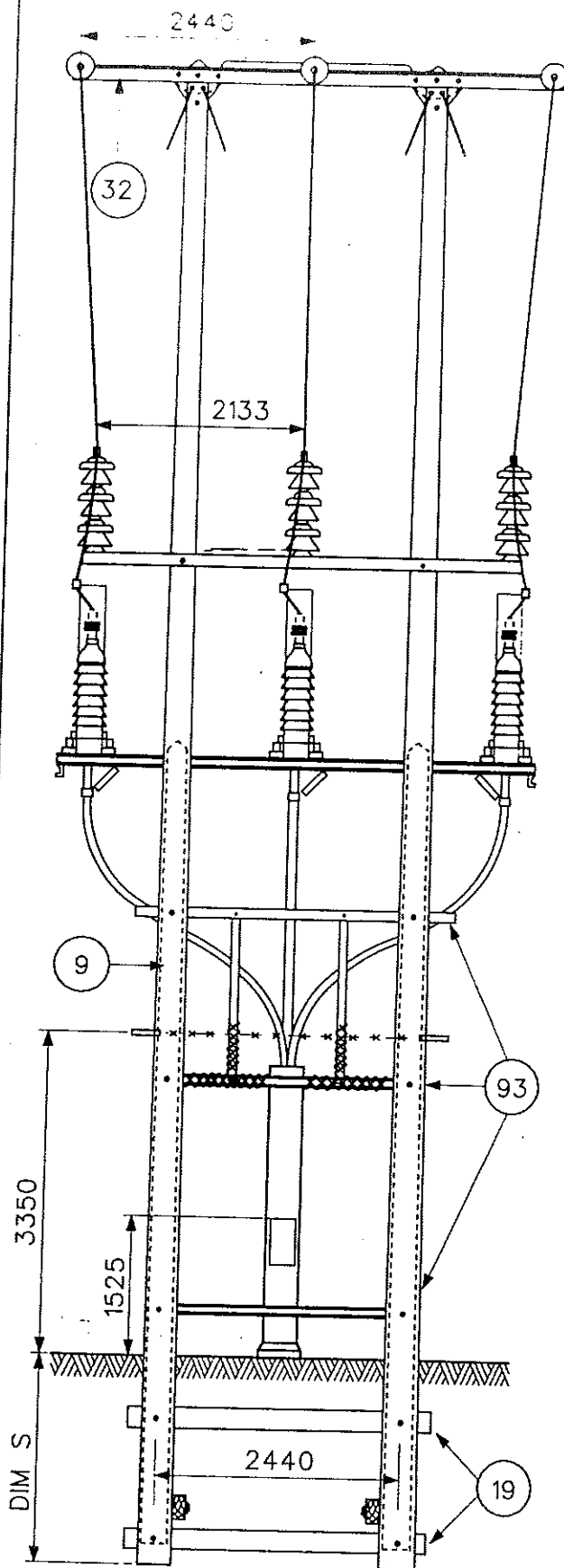
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MATERIALS LIST.

Item	Nº	Commodity Code	Description	Drawing or Ref.Number	M 20 Bolts	Washers		
						sc	sf	r
9	2	BPC 024	Wood Pole	Y204 L 0604				
16	4	217539	Kicking/Stay Block (type 2)	439103	4(Pole)	4	4	
19	2	217497	Brace Block	Y204 L 0301	4(Pole)	4	4	
32	1	223669	Crossarm Member	Y204 L 0904	4 x 300			
57	3	215970	66kV Pilot Insulator	Y204 L 0702				
60	3		33kV Tension Set	Y204 L 0704				
			66kV Tension Set					
70	4	294181	Safety Sign	Y201 L 1002				
71			Operational Plates	Y201 L 1001/7				
73	4		Stay Arrangement	Y204 L 0501				
74	1	224519	Anti- Climbing Guard	Y201 L 1101/2				
80	3		66kV Sealing End Connector	Y204 L 0801				
82	3	221119	Surge Diverter Connector	Y204 L 0803				
83	3	221093	33kV Post Ins.Clamp(al)	Y204 L 0804				
		221101	66kV Post Ins.Clamp(cu)	Y204 L 0805				
92	1		66kV Sealing End P-form (15 - 18m poles)	Y204 L 0909	10(Pole)			
93	1		66kv Cable Supports	Y204 L 0910	6(Pole)			
94	1	224535	Cable Guard	Y204 L 0911	4 x 50			
95	1		Link Box	Eng Rec C55				

* 4 off M20 x 60 bolts also required for gusset plates.

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DIMENSIONS IN MILLIMETRES

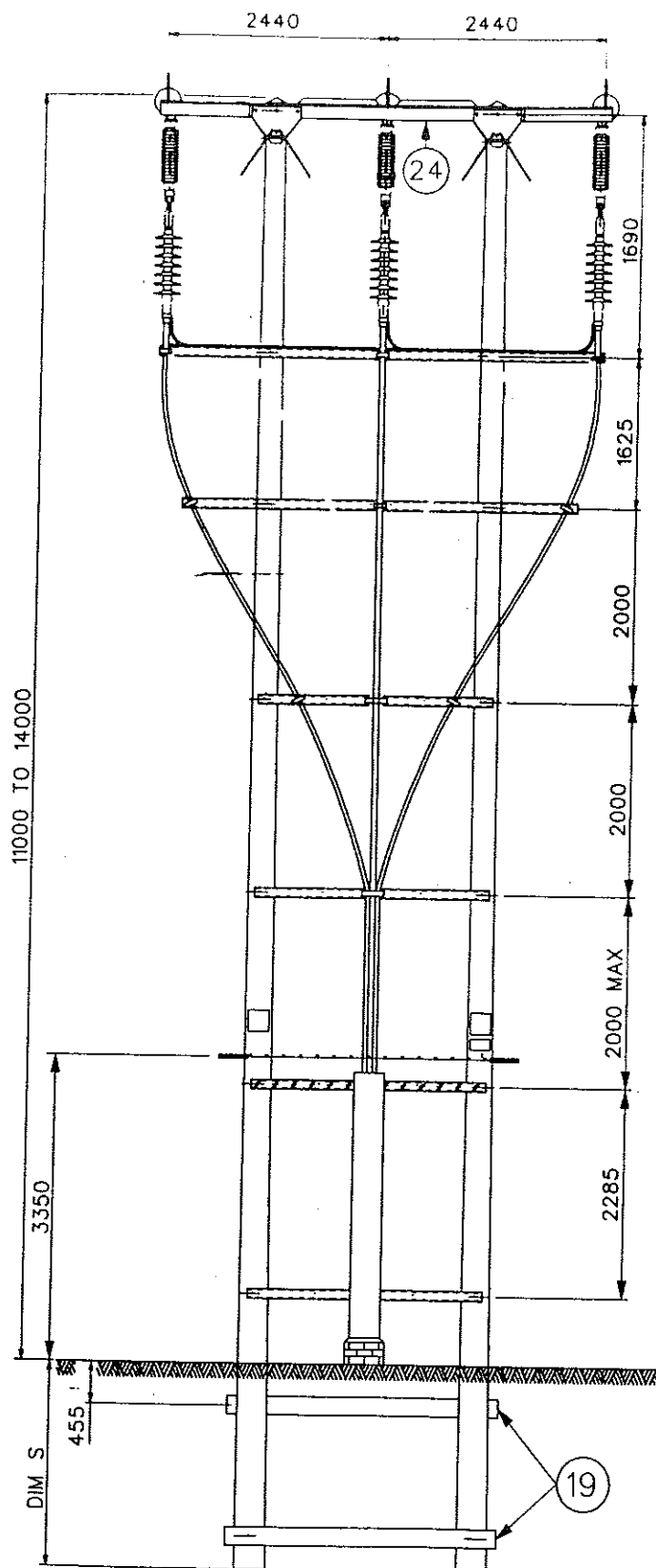
drawn	BT	CE/C/37 CE/OM/DC1 66kV CABLE TERMINATION WITH SURGE DIVERTERS 15.0 TO 18.0m POLES		 Yorkshire Electricity
traced	SW			
checked	JMG			
approved	BT			
revision	date 28-3-91	original scale	Y104L0108	

MATERIALS LIST.

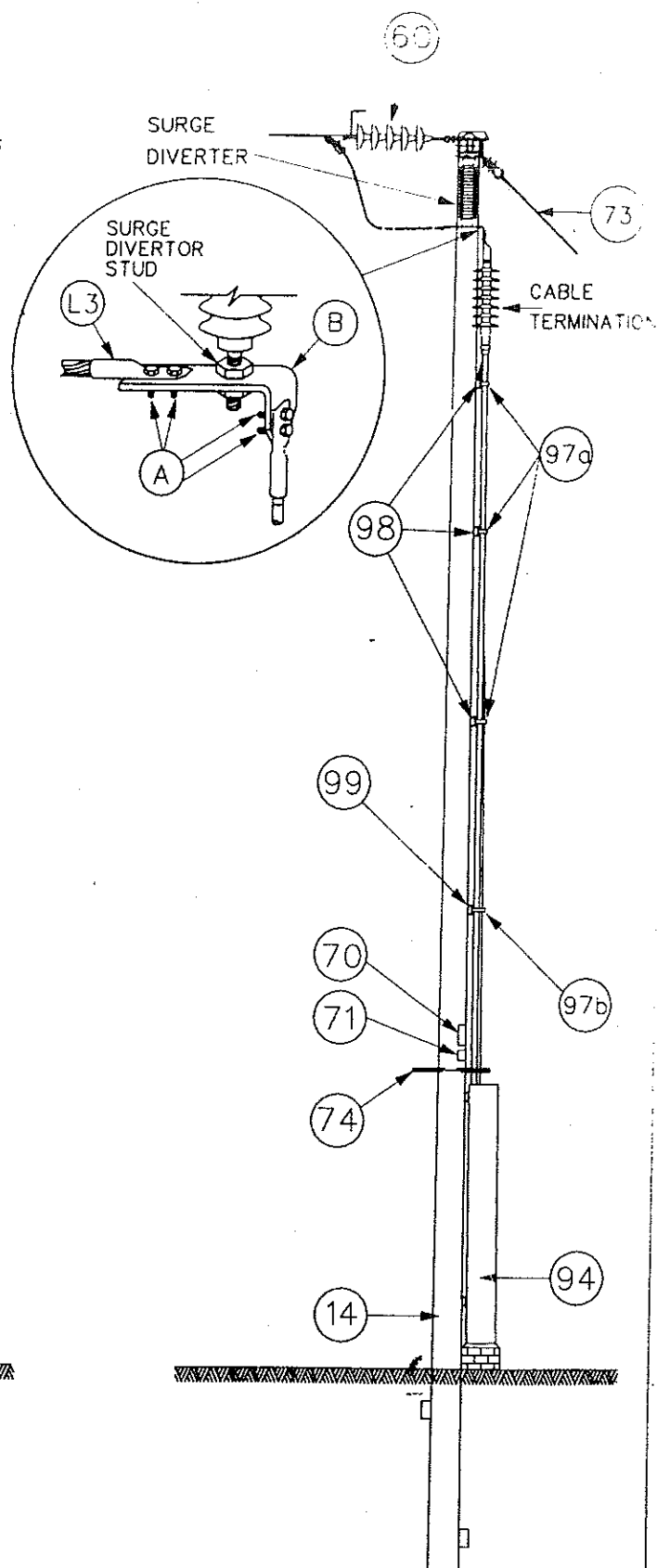
Item	Nº	Commodity Code	Description	Drawing or Ref.Number	M 20 Bolts	Washers		
						sc	sf	r
14	2	BPC 024	Wood Pole	Y204 L 0605				
19	2	217497	Brace Block	Y204 L 0301	4(Pole)	4	4	
24	1	223669	Crossarm Member	Y204 L 0904	4 x 300*			
60	3		33kV Tension Set	Y204 L 0704				
70	2	294181	Safety Sign	Y201 L 1002				
71			Operational Plates	Y201 L 1001/7				
73	4		Stay Arrangement	Y204 L 0501				
74	1	224519	Anti- Climbing Guard	Y201 L 1101/2				
98	1		33kV Cable Support Steelwork	Y204 L 0907	10(Pole)	10		
94	1	224535	Cable Guard	Y204 L 0911	4 x 50			
97	9		33kV Cable Cleats Single					
97	3		33kV Cable Cleats Trefoil					
L3	3		Straight Lug 95/125mm ²					
A	12		2 x 12mm Stud					
B	3		M14 x 60 bolts					
			Termination Plate	Y 204 L 0803				

* 8 off M20 x 45 bolts also required for gusset plates.

Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=11.5	1.80
12.0 - 13.0	2.00
14.0 - 15.0	2.10
16.0 - 17.0	2.25
18.0 - 20.0	2.40



ELEVATION



SIDE ELEVATION

DIMENSIONS IN MILLIMETRES

drawn

G.R.
12.5.94

traced

S.W.
12.5.94

checked

—

approved

C.N.

revision

—

date

18.3.98

original scale

NTS

Y104L0109

CAD

33kV POLYMERIC CABLE TERMINATION

WITH POLYMERIC TYPE HEC36 SURGE DIVERTERS

CE/C/37 (M)

Y/E

Yorkshire Electricity



Yorkshire Electricity

MATERIALS LIST.

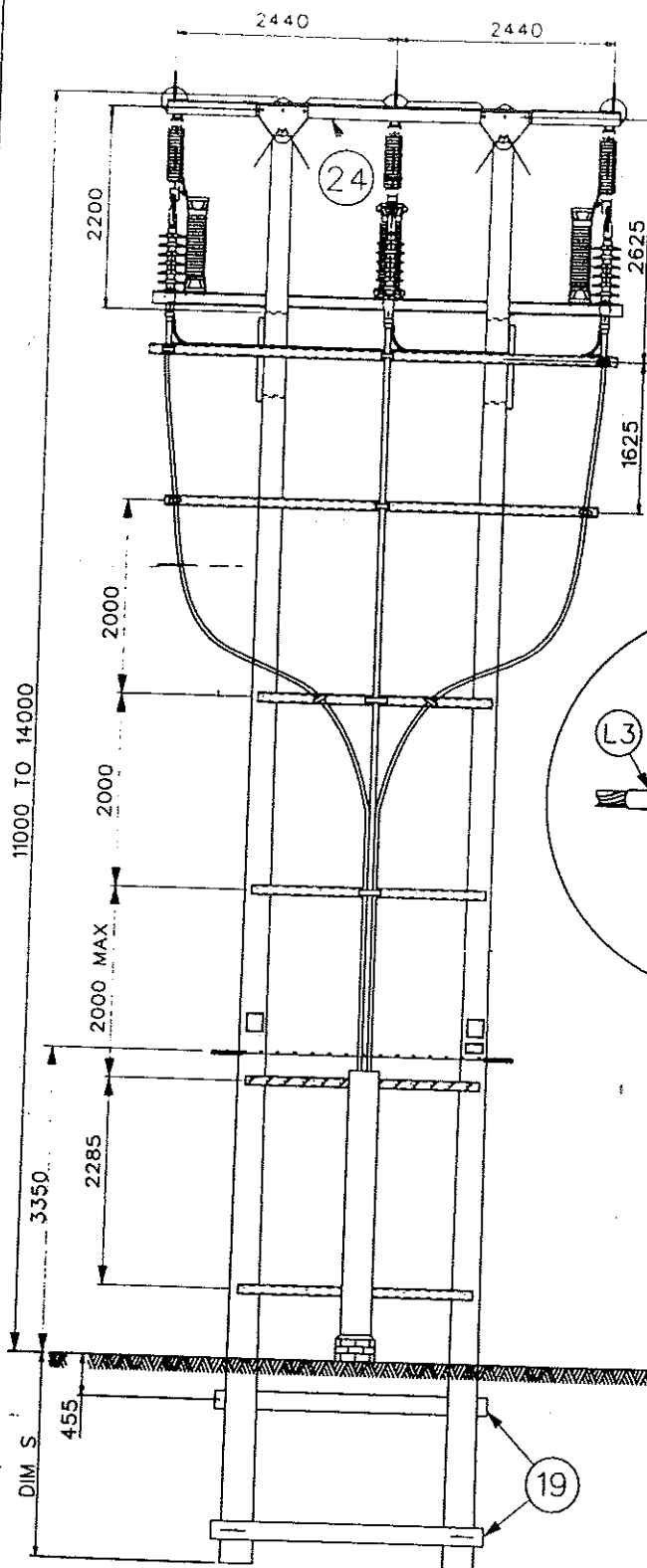
Item	Nº	Commodity Code	Description	Drawing or Ref. Number	M 20 Bolts	Washers		
						sc	sf	r
6	2		Wood Pole	Y 204 L 0606				
19	2	217497	Brace Block	Y204 L 0301	4(Pole)	4	4	
24	1	223669	Crossarm Member	Y204 L 0904	4 x 300			
60	3		33kV Tension Set	Y204 L 0704				
			66kV Tension Set					
70	4	294181	Safety Sign	Y201 L 1002				
71			Operational Plates	Y201 L 1001/7				
73	4		Stay Arrangement	Y204 L 0501				
74	1	224519	Anti- Climbing Guard	Y201 L 1101/2				
88	1	228395	Sealing End P-Form	Y204 L				
97a	9		33kV Cable Support (Single)	Y204 L 0915				
97b	1		33kV Cable Support (Trefoil)	Y204 L 0915				
98	9		Cable Support	Y 204 L 0917				
99	1		Cable Support	Y 204 L 0917				
95	1		Link Box	Eng Rec C55				
L3	3		Straight Lug 95/125mm ²					
			2 x 12mm Stud					
A	12		M14 x 60 bolts					
B	3		Termination Plate	Y 204 L 0803				
94	1	224535	Cable Guard	Y204 L 0911	4 x 50			

* 8 off M20 x 45 bolts also required for gusset plates.

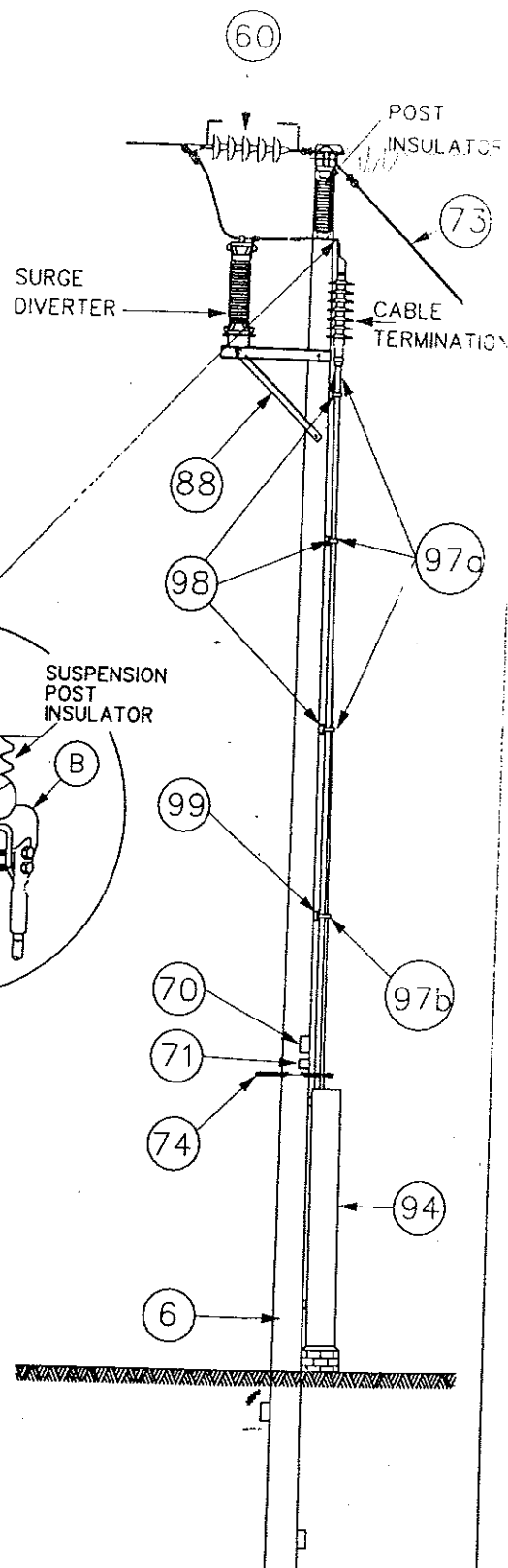
Pole Planting Depth	
Pole Length (m)	Dim S (m)
<=11.5	1.80
12.0 - 13.0	2.00
14.0 - 15.0	2.10
16.0 - 17.0	2.25
18.0 - 20.0	2.40

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 File: _____
 Plotted on: _____



ELEVATION



SIDE ELEVATION

DIMENSIONS IN MILLIMETRES

drawn	G.R.
	12.5.94
traced	S.W.
	12.5.94
checked	-
approved	C.N.
revision	A

66kV POLYMERIC CABLE TERMINATION
 WITH PORCELAIN TYPE MB75 SURGE DIVERTERS
 CE/C/37 (M) - CE/OM/DC1

date 18.3.98

original
 scale NTS

Y104L0111

CAD



Yorkshire
 Electricity